DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

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PROJECT TITLE:

EFFECTS OF CONCRETE COVER AND WATER-CEMENT RATIO ON HYDROGEN GENERATION RISK IN REINFORCED CONCRETE STRUCTURES

A Machine Learning-Assisted Experimental Investigation

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A research proposal submitted in partial fulfilment of the requirement for the award of the degree of Bachelor of Education Technology in Civil Engineering of Dedan Kimathi University of Technology.

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DECLARATION

We, Kanjiru Newton Mogaka and Kamau Vivian Wangari, hereby declare that this research proposal is our original work and has not been submitted for any degree award in any other university or institution. All citations from other people's work have been duly identified and acknowledged within the list of references.

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This research proposal has been submitted for examination with my approval as the University supervisor.

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DEDICATION

This work is dedicated to our families, whose support has made this journey possible, and to everyone striving to make construction in Kenya safer and more durable.

ACKNOWLEDGEMENTS

We would like to acknowledge the Almighty God for guidance throughout this research process. We are grateful to Dedan Kimathi University of Technology for the opportunity to pursue this research. Special thanks go to [Supervisor's Name], our project supervisor, for the guidance, patience, and constructive feedback offered throughout the development of this proposal. We also acknowledge the Department of Civil Engineering for the technical support and laboratory access extended to this study. Finally, we thank our family and friends for their continued encouragement.

ABSTRACT

Reinforced concrete durability in Kenya and the wider East African region is governed largely by durability provisions developed from research in temperate, European exposure conditions. One failure mode that has received limited local attention is hydrogen-induced stress, whereby atomic hydrogen produced as a byproduct of corrosion at the rebar-concrete interface diffuses into the steel lattice and reduces its resistance to brittle fracture. This proposal sets out an experimental investigation into how two of the most fundamental durability design variables, concrete cover depth and water-cement (w/c) ratio, influence the risk of hydrogen generation in reinforced concrete exposed to accelerated chloride ingress. Thirty-six reinforced concrete prism specimens will be cast using Kenyan-sourced cement, aggregates and Grade 410 mild steel rebar, spanning four cover depths (20, 30, 40 and 50 mm) and four w/c ratios (0.40, 0.45, 0.50 and 0.55), and will be subjected to eight weeks of sodium chloride (NaCl) ponding. Half-cell potential and linear polarisation resistance (LPR) measurements will be recorded weekly, and corrosion current density will be converted into a hydrogen generation rate proxy using Faraday's law. The resulting dataset will be analysed with a Random Forest model supported by SHAP (SHapley Additive exPlanations) feature importance analysis to determine which of the two variables, cover depth or w/c ratio, is the dominant predictor of hydrogen generation risk, and to characterise how the two variables interact. Findings will be benchmarked against the Life-365, Bamforth, and Andrade-Alonso models to establish whether internationally calibrated durability provisions are adequate for Kenya's coastal and industrial exposure conditions. The study is expected to yield the first Kenyan experimental dataset linking cover depth and w/c ratio to hydrogen generation risk, together with practical recommendations for minimum cover and maximum w/c specifications under BS 8500 exposure classes XS and XD.

TABLE OF CONTENTS

LIST OF TABLES

(Additional tables of test results, e.g. weekly half-cell potential and i_corr results, will be added following data collection in Phase 3.)

LIST OF FIGURES

(Additional figures, e.g. SHAP dependence plots and response surface maps, will be added following data analysis in Phase 5.)

LIST OF ACRONYMS AND ABBREVIATIONS

RC - Reinforced Concrete

w/c - Water-Cement Ratio

HE - Hydrogen Embrittlement

i_corr - Corrosion Current Density

LPR - Linear Polarisation Resistance

SHAP - SHapley Additive exPlanations

ML - Machine Learning

NaCl - Sodium Chloride

OPC - Ordinary Portland Cement

KS EAS - Kenya Standard, East African Standard

BS EN - British Standard, European Norm

fib - International Federation for Structural Concrete

DeKUT - Dedan Kimathi University of Technology

ASTM - American Society for Testing and Materials

CHAPTER 1: INTRODUCTION

1.1 Background Information

Reinforced concrete (RC) is the dominant structural material in Kenya and across East Africa, forming the backbone of residential, commercial, infrastructure, and industrial construction. The durability and long-term structural integrity of RC structures depend critically on the condition of the embedded steel reinforcement. Concrete cover, the layer of concrete between the outer surface of the structure and the reinforcing steel, acts as a physical barrier against the ingress of deleterious agents such as chlorides, carbon dioxide, sulphates, moisture and oxygen, all of which can trigger or sustain reinforcement corrosion (Kenya Engineer, 2024).

One of the most insidious yet underexamined threats to RC durability is hydrogen-induced stress, a mechanism by which atomic hydrogen generated during rebar corrosion diffuses into the steel lattice, causing embrittlement and, in severe cases, brittle fracture at stress levels well below the nominal yield strength. When the passive oxide film on the rebar surface is broken down, typically through chloride ingress or carbonation of the concrete cover, an electrochemical corrosion cell forms on the steel surface. The cathodic half-reaction of this cell produces nascent atomic hydrogen, which can either combine into harmless hydrogen gas or, under certain conditions, diffuse directly into the steel microstructure before recombination occurs (Tuutti, 1982). Once inside the steel, hydrogen accumulates at grain boundaries, inclusions and crack tips, reducing the cohesive bond energy of the lattice and promoting brittle fracture, a phenomenon that has been documented extensively in laboratory studies of cathodically polarised and corroding steel but has received little attention in the context of tropical, East African exposure conditions.

Kenya's coastal belt, including Mombasa, Malindi and Lamu, presents chloride-laden marine environments in which airborne and waterborne chlorides, combined with high ambient temperature and relative humidity, sustain aggressive reinforcement corrosion once it has been initiated (Kenya Engineer, 2024). Inland industrial zones such as Athi River, Thika and Ruiru expose structures to aggressive chemical atmospheres associated with carbonation-induced corrosion. Despite these conditions, no experimental study appears to have characterised the relationship between corrosion and hydrogen generation using locally available cement, aggregates and rebar under Kenyan ambient conditions. Existing local research on corrosion in Kenyan RC structures has largely focused on non-destructive assessment of existing structures rather than the systematic, laboratory-controlled variation of design parameters such as cover depth and w/c ratio.

This gap represents both a research opportunity and a practical risk. Structural engineers in Kenya routinely apply durability provisions embedded in codes and service life models, such as Life-365 and the fib Bulletin 34 (DuraCrete) framework, that were calibrated using data from temperate climates. Given that elevated ambient temperature is known to accelerate chloride diffusion and corrosion initiation, there is a reasonable possibility that these internationally calibrated provisions understate the durability risk, including hydrogen-related embrittlement risk, faced by structures in Kenya's coastal and industrial exposure classes.

1.2 Problem Statement

The relationship between concrete cover depth, water-cement (w/c) ratio, and the risk of hydrogen generation at the rebar-concrete interface is well established in global literature, generally through its role in controlling the onset and rate of corrosion (Bamforth, 1999; Andrade & Alonso, 2004). However, this relationship has not been experimentally characterised using Kenyan construction materials and under Kenyan ambient conditions. Several key unknowns motivate this study:

- Whether chloride diffusion coefficients derived from local Portland cement brands (such as Bamburi, Savannah, and Mombasa Cement) align with the global benchmarks embedded in service life models such as Life-365 and the fib DuraCrete/Bulletin 34 framework.
- Whether the accelerated corrosion initiation timelines expected under Kenya's higher mean ambient temperatures, relative to the European baseline used in most global models, translate into proportionally higher hydrogen generation rates.
- Which of the two primary design variables, concrete cover depth or w/c ratio, exerts the greater influence on hydrogen generation risk, and how the two variables interact, a question with direct implications for cost-optimised durable design in Kenyan practice.
- Whether existing Kenyan structural standards, including KS EAS 18 (which references BS 8500/EN 206), provide adequate durability provisions for hydrogen-related embrittlement risk in the exposure classes prevalent locally.

This study addresses these gaps through a structured laboratory experiment designed to quantify hydrogen generation risk as a function of cover depth and w/c ratio, followed by a machine learning-assisted sensitivity analysis to identify the dominant design variable and characterise the interaction surface between the two parameters.

1.3 Research Objectives

1.3.1 Overall Objective

To experimentally determine the effects of concrete cover depth and water-cement ratio on hydrogen generation risk in reinforced concrete specimens under accelerated chloride exposure, using electrochemical measurements as a proxy for hydrogen evolution rate.

1.3.2 Specific Objectives

- SO1: To cast and cure 48 reinforced concrete prism specimens with systematically varied cover depths (20 mm, 30 mm, 40 mm, 50 mm) and w/c ratios (0.40, 0.45, 0.50, 0.55) using Kenyan-sourced materials.

- SO2: To subject the specimens to accelerated chloride ingress via NaCl ponding and monitor corrosion initiation and progression through half-cell potential and linear polarisation resistance (LPR) measurements over an eight-week exposure period.

- SO3: To derive hydrogen generation rate proxies from measured corrosion current densities using Faraday's law and construct a structured experimental dataset.

- SO4: To apply a machine learning model (Random Forest with SHAP value analysis) to the experimental dataset to quantify the relative influence of cover depth versus w/c ratio on hydrogen generation risk and characterise their interaction effect.

- SO5: To compare experimental findings against global benchmarks (the Andrade-Alonso classification, Bamforth cover-depth relationships, and Life-365 diffusion model predictions) and identify deviations attributable to local material and climatic conditions.

1.4 Research Questions

- How does concrete cover depth influence the rate of corrosion-induced hydrogen generation in RC specimens fabricated with Kenyan construction materials under accelerated chloride exposure?

- How does water-cement ratio independently affect hydrogen generation risk, and does its influence supersede or interact with that of cover depth?
- Do experimentally derived chloride diffusion coefficients and corrosion initiation timelines align with the predictions of globally calibrated service life models?
- Which design variable, cover depth or w/c ratio, is the dominant predictor of hydrogen generation risk as identified by machine learning feature importance analysis?

1.5 Scope and Limitations of the Study

1.5.1 Scope of the Study

The study is scoped to mild steel rebar (Grade 410) representative of standard Kenyan construction practice. High-strength prestressing steel, which exhibits greater susceptibility to hydrogen embrittlement, falls outside the scope of this study. The experimental programme covers four cover depths (20-50 mm) and four w/c ratios (0.40-0.55), reflecting the range commonly encountered in Kenyan residential and commercial construction. The chloride exposure regime is limited to accelerated NaCl ponding rather than long-term natural marine exposure, since the latter would exceed the timeframe available for an undergraduate research project.

1.5.2 Limitations of the Study

Hydrogen generation risk is inferred from electrochemical proxies, specifically corrosion current density converted via Faraday's law, rather than measured directly through hydrogen concentration testing, which would require specialised permeation cell equipment unavailable at undergraduate level. The machine learning component is framed as an exploratory sensitivity analysis rather than a deployable predictive model, given the dataset size constraints inherent in a single-study experiment of this scale. Accelerated laboratory exposure conditions, while necessary to obtain results within the project timeline, cannot fully replicate the variability of field exposure, including wetting-drying cycles, wind-driven salt spray, and seasonal temperature variation experienced by real structures in coastal Kenya. These limitations are acknowledged transparently and do not diminish the validity of the comparative and analytical contributions of the study.

1.6 Significance and Justification of the Study

1.6.1 Significance of the Study

This study makes a contribution at three levels simultaneously. Academically, it generates what appears to be the first experimentally derived dataset characterising the cover depth-w/c ratio-hydrogen generation relationship using East African construction materials, and it demonstrates a methodological approach combining electrochemical experimentation with machine learning feature importance analysis that is applicable to broader durability research. From a professional practice standpoint, comparing local experimental results against global model predictions provides evidence-based guidance on whether international durability code provisions are conservative, adequate, or insufficient for Kenyan exposure conditions, directly informing the work of structural engineers specifying cover and mix design for durable RC construction. From a policy and standards perspective, the findings provide a foundation for future advocacy toward explicitly incorporating hydrogen embrittlement risk as a distinct consideration in Kenyan structural standards, particularly for coastal and industrial exposure classes.

1.6.2 Justification of the Study

Given that RC remains the material of choice for the majority of Kenyan construction, and that the coastal and industrial belts represent some of the country's most economically important zones for infrastructure development, a locally grounded understanding of the interaction between cover depth, w/c ratio and hydrogen-related embrittlement risk is of direct practical value. Where existing specifications may understate this risk, the cost of over-reliance on internationally calibrated code provisions could be measured in reduced service life and increased maintenance and repair costs for Kenyan RC infrastructure.

CHAPTER 2: LITERATURE REVIEW

2.1 Overview

This chapter reviews existing literature on the corrosion of steel in reinforced concrete, the mechanism of hydrogen generation and hydrogen-induced stress, the roles of concrete cover depth and water-cement ratio in controlling durability risk, the electrochemical techniques used to monitor corrosion, the global service life prediction models against which local results will be benchmarked, and the growing use of machine learning and SHAP-based interpretability methods in concrete durability research. The chapter closes by situating these strands of literature within the Kenyan and East African context and identifying the specific research gap addressed by this study.

2.2 Corrosion of Steel in Reinforced Concrete

Steel embedded in concrete is normally protected from corrosion by a thin, adherent passive oxide film that forms and is sustained by the highly alkaline pore solution of hydrating cement. Corrosion begins once this passive film is broken down, a process referred to as depassivation, which is caused primarily by two mechanisms: chloride ingress and carbonation of the concrete cover (Tuutti, 1982). Once depassivation occurs, an electrochemical corrosion cell is established on the steel surface, comprising anodic sites where iron is oxidised and cathodic sites where oxygen and water are reduced, generating hydroxide ions and, under certain conditions, nascent atomic hydrogen.

2.2.1 Chloride-Induced Corrosion

Chloride-induced corrosion is generally considered more severe and more localised than carbonation-induced corrosion, and is typically associated with marine environments, coastal regions, and structures exposed to de-icing salts (Current Problems in Research, 2025). Unlike carbonation, which causes a broadly uniform loss of alkalinity across the cover depth, chloride ions disrupt the passive film locally without significantly reducing the overall pH of the surrounding concrete, producing pitting-type corrosion that can proceed rapidly once initiated. Chloride penetration into concrete is most commonly modelled using Fick's second law of diffusion, and diffusion coefficients are known to vary considerably with mix design, curing regime, and the presence of supplementary cementitious materials (Liang & Wang, 2021).

2.2.2 Carbonation-Induced Corrosion

Carbonation-induced corrosion results from the reaction of atmospheric carbon dioxide with calcium hydroxide in the cement paste, progressively lowering the pH of the pore solution until the passive film is no longer thermodynamically stable. This mechanism is more common in inland and industrial regions, including car parks and structures away from direct marine exposure, and its prediction is complicated by the fact that it involves a chemical reaction front rather than a pure diffusion process (Kenya Engineer, 2024).

In the Kenyan context, both mechanisms are relevant: chloride-induced corrosion dominates in coastal counties such as Mombasa, Kilifi and Lamu, while carbonation-induced corrosion is more relevant to industrial zones such as Athi River, Thika and Ruiru.

2.3 Hydrogen Generation and Hydrogen-Induced Stress in Reinforcing Steel

2.3.1 Mechanism of Hydrogen Embrittlement

Hydrogen embrittlement (HE) refers to the loss of ductility and the onset of brittle fracture in a metal caused by the absorption of atomic hydrogen into its lattice. During the corrosion of steel in concrete, the cathodic half-reaction can generate atomic hydrogen at the steel surface before it recombines into molecular hydrogen gas; while most of this hydrogen escapes as gas, a portion diffuses into the steel microstructure, particularly at regions of high triaxial stress or lattice defects such as grain boundaries and inclusions (RILEM Technical Committee 293-CCH, n.d.). Once absorbed, hydrogen reduces the cohesive energy between metal atoms and can promote sudden brittle fracture at applied stresses well below the nominal yield strength of the steel, a failure mode that by its nature gives little visible warning before failure.

2.3.2 Hydrogen Generation as a Byproduct of Cathodic Reactions

Experimental work applying controlled cathodic potentials to steel immersed in simulated carbonated and chloride-contaminated concrete pore solutions has shown that hydrogen generation and subsequent embrittlement can be induced and measured under laboratory conditions, and that the degree of embrittlement is closely tied to the applied cathodic overpotential and the resulting hydrogen concentration absorbed into the steel (ScienceDirect, 2015). Related studies on cathodically protected reinforced concrete structures in marine environments have similarly found that even modest cathodic current densities can be sufficient to deplete dissolved oxygen at the steel-concrete interface and generate measurable hydrogen production once the local pH has been lowered by corrosion-related processes. These findings support the use of corrosion current density, a quantity that is comparatively straightforward to measure electrochemically, as a proxy for the rate of hydrogen generation at the rebar surface, since both processes share the same underlying cathodic reaction.

2.4 Influence of Concrete Cover Depth on Corrosion and Hydrogen Generation Risk

Concrete cover functions as the primary physical barrier limiting the ingress of chlorides, carbon dioxide, moisture and oxygen to the level of the reinforcement, and its depth is one of the most direct design controls available for durability. Field surveys combined with laboratory investigation have found that a cover depth of over 40 mm can meaningfully reduce the risk of corrosion-induced spalling, even where the cover concrete itself is relatively porous or has a comparatively high w/c ratio, whereas structures with less than 40 mm of cover combined with a supply of water were found to experience more severe corrosion-induced spalling (PMC, 2021). The same body of work found that maximum moisture evaporation, and therefore maximum oxygen availability at the steel surface, tended to occur at cover depths of approximately 30-40 mm under typical drying conditions, suggesting that the relationship between cover depth and corrosion risk is not perfectly linear and may involve a transition zone rather than a single threshold.

Because hydrogen generation at the steel surface is tied to the same cathodic reaction that drives corrosion, cover depth is expected to influence hydrogen generation risk indirectly, by controlling the time to depassivation and the availability of oxygen and moisture at the cathode. This relationship, however, does not appear to have been quantified directly in existing literature, which has instead focused on cover depth's role in corrosion initiation and cover cracking rather than on hydrogen-specific outcomes.

2.5 Influence of Water-Cement Ratio on Chloride Ingress and Corrosion

The water-cement ratio governs the porosity and pore connectivity of the hardened cement paste, and by extension the ease with which chlorides, moisture and oxygen can move through the cover concrete to reach the reinforcement. Studies examining chloride penetration across a range of w/c ratios have generally found that chloride ingress increases with increasing w/c ratio, with correlations reported across w/c ratios spanning roughly 0.40 to 0.75 in some studies (Academia.edu, 2025). Comparable work modelling the full-service life of reinforced concrete in chloride environments has shown that w/c ratio affects both the chloride penetration stage, governed by Fick's second law, and the subsequent corrosion damage stage, meaning that its influence compounds across the service life of a structure rather than being confined to the initiation period alone (Liang & Wang, 2021).

Some studies have also reported an inverse relationship in specific circumstances, for example where high fly ash replacement is combined with a higher nominal w/c ratio, illustrating that w/c ratio does not act in isolation and that its effect on durability can be modified by the presence of supplementary cementitious materials (ScienceDirect, 2025). For the plain Portland cement mixes considered in this study, however, the general expectation from the literature is that hydrogen generation risk, being downstream of corrosion current density, should increase with increasing w/c ratio, consistent with the broader literature linking higher w/c ratios to greater carbonation depth and reduced resistance to chloride ingress (Current Problems in Research, 2025).

2.6 Electrochemical Techniques for Corrosion and Hydrogen Generation Assessment

2.6.1 Half-Cell Potential Method

The half-cell potential method, standardised in ASTM C876, measures the electrochemical potential difference between the reinforcing steel and a reference electrode, typically a copper-copper sulphate electrode, placed on the concrete surface. Potentials more negative than -350 mV relative to the copper-copper sulphate electrode are generally associated with a greater than 90 percent probability of active corrosion, while potentials less negative than -200 mV are associated with a less than 10 percent probability, with readings in between considered inconclusive (Giatec Scientific, 2026). The method is valuable for locating areas of elevated corrosion risk but is explicitly qualitative: it indicates the probability that corrosion is occurring rather than the rate at which it is proceeding, and its readings can be influenced by cover depth, concrete resistivity, moisture state and temperature (Assouli et al., 2008).

2.6.2 Linear Polarisation Resistance

Linear polarisation resistance (LPR) testing complements half-cell potential measurement by providing a quantitative estimate of corrosion current density, generally denoted i_corr, expressed in microamps per square centimetre. Corrosion current densities below roughly 1 microamp per square centimetre are typically considered indicative of low corrosion activity, while values above roughly 10 microamps per square centimetre are considered severe (Giatec Scientific, 2026). Because i_corr is directly related to the rate of metal dissolution at the anode and, by the stoichiometry of the corrosion cell, to the rate of the complementary cathodic reaction, it provides a practical basis for estimating hydrogen generation rate without requiring direct hydrogen concentration measurement.

2.6.3 Faraday's Law and Corrosion Current Density

Faraday's law of electrolysis relates the mass of a substance produced or consumed at an electrode to the total electrical charge passed through the electrochemical cell. Applied to a corroding rebar, Faraday's law allows the corrosion current density measured via LPR to be converted into an equivalent rate of metal loss, and by extension, given known stoichiometry of the cathodic hydrogen evolution reaction, into an equivalent proxy rate of hydrogen generation. This approach has been used in integrated service life models that combine Faraday's electrolysis principle with Fick's diffusion law to forecast chloride-induced corrosion service life, with reported gains in prediction reliability when additional factors such as humidity and concrete resistivity are incorporated (Tandfonline, 2025).

2.7 Global Service Life and Durability Prediction Models

2.7.1 The Life-365 Model

The Life-365 model is a widely used software tool for predicting the service life of reinforced concrete exposed to chlorides, developed originally through an industry consortium and now maintained under ACI Committee 365 (Life-365 Consortium, 2013). The model assumes chloride ingress is governed by Fick's second law of diffusion under given assumptions about surface chloride concentration and its build-up over time, and calculates an initiation period, the time required for chlorides to penetrate the cover and reach a critical threshold concentration at the level of the reinforcement, as well as a subsequent propagation period leading to the end of service life. Cover depth and the chloride diffusion coefficient of the specific concrete mix, itself strongly influenced by w/c ratio, are two of the most consequential inputs to the model.

2.7.2 The fib Bulletin 34 / DuraCrete Framework

The fib Bulletin 34 Model Code for Service Life Design, building on the earlier DuraCrete research programme, provides a probabilistic framework for durability design that similarly treats chloride diffusion as the governing transport mechanism during the initiation period, but places greater emphasis on partial safety factors and reliability-based design than the deterministic Life-365 approach (fib Bulletin 34, 2006).

Both frameworks were developed and calibrated predominantly using data from temperate European and North American exposure conditions, which is the basis for this study's concern that their default parameters may not transfer directly to Kenyan ambient conditions.

2.7.3 Bamforth Cover-Depth Relationships

Bamforth's analysis of long-term UK coastal exposure trials established empirical relationships between cover depth, concrete quality and the time to corrosion initiation, providing some of the input data subsequently embedded in service life models such as Life-365 (Bamforth, 1999). These relationships remain widely referenced in durability specification but were derived from UK marine exposure conditions with a substantially lower mean ambient temperature than Kenya's coastal belt.

2.7.4 Andrade-Alonso Corrosion Rate Classification

Andrade and Alonso proposed a classification of corrosion rate ranges, derived from polarisation resistance measurements, that is widely used to categorise the severity of ongoing corrosion in field and laboratory studies, and which this study will use as a reference scale against which measured i_corr values from the Kenyan specimens can be compared (Andrade & Alonso, 2004).

2.8 Machine Learning and SHAP-Based Feature Importance in Concrete Durability Research

Machine learning methods, particularly ensemble tree-based models such as Random Forest, Extra Trees, and gradient boosting variants, have become increasingly common in concrete materials research for predicting properties such as compressive strength, abrasion resistance and durability indicators from mix design and exposure variables (PMC, 2025). A recurring limitation of these models is their limited interpretability relative to simpler statistical models: while they can achieve high predictive accuracy, it is often unclear which input variables are driving a given prediction and in what direction.

SHAP (SHapley Additive exPlanations), introduced by Lundberg and Lee (2017), addresses this limitation by adapting the concept of Shapley values from cooperative game theory to machine learning, treating each input feature as a player contributing to the model's prediction and distributing credit for that prediction fairly among the features. For tree-based models such as Random Forest, an efficient variant known as TreeSHAP allows Shapley values to be computed exactly and quickly, making SHAP practical for datasets of the scale expected in this study (Interpretable Machine Learning, n.d.). In concrete-related applications, SHAP analysis has been used to identify cement content, water content and admixture dosage as dominant predictors of compressive strength, and to reveal interaction effects between water-to-binder ratio and other mix parameters that would not be visible from a simple feature-ranking exercise (ScienceDirect, 2025). This study applies the same logic to a durability outcome, using SHAP to determine whether cover depth or w/c ratio dominates hydrogen generation risk and how the two interact, rather than to a strength outcome as in most prior concrete-related SHAP studies.

2.9 Reinforced Concrete Durability in the Kenyan and East African Context

Existing Kenyan literature on RC corrosion has concentrated on the assessment of existing structures rather than the controlled experimental variation of design parameters. Non-destructive testing studies combining half-cell potential, carbonation depth testing and chloride testing have highlighted the inadequacy of relying on visual inspection alone to assess the structural integrity of Kenyan concrete structures and have confirmed that chloride-induced corrosion is a significant concern for coastal structures requiring regular inspection (Academia.edu, 2014). Separately, industry commentary has emphasised that concrete cover depth, quality and condition are seldom given sufficiently rigorous consideration in day-to-day Kenyan design practice, despite their acknowledged importance to durability outcomes, and has noted that Mombasa's combination of airborne chlorides, high ambient temperature and high relative humidity is particularly conducive to sustaining corrosion propagation once initiated (Kenya Engineer, 2024).

What is absent from this body of local work is any experimental dataset directly relating cover depth and w/c ratio, under controlled laboratory conditions using Kenyan materials, to a corrosion-derived hydrogen generation outcome, nor any comparison of Kenyan chloride diffusion behaviour against the parameters embedded in Life-365, Bamforth or Andrade-Alonso reference models.

2.10 Research Gap

The reviewed literature confirms that the mechanisms linking cover depth, w/c ratio, corrosion and hydrogen generation are individually well understood, and that electrochemical methods (half-cell potential, LPR, Faraday's law) and machine learning interpretability tools (Random Forest, SHAP) are individually well established. What has not been done, based on the literature reviewed, is the integration of these elements into a single, locally grounded experimental study: casting specimens with Kenyan cement, aggregates and rebar; systematically varying cover depth and w/c ratio; deriving a hydrogen generation proxy via Faraday's law from measured i_corr; and applying SHAP-based Random Forest analysis to determine the dominant design variable and its interaction effects, benchmarked explicitly against Life-365, Bamforth and Andrade-Alonso reference values. This study is positioned to fill that gap.

CHAPTER 3: MATERIALS AND RESEARCH METHODOLOGY

3.1 Introduction

This chapter describes the materials, experimental design, specimen preparation, testing procedures and data analysis methods to be used in this study. The methodology is structured around a 4 (cover depth) x 4 (w/c ratio) factorial experiment with three replicates per combination, giving 48 reinforced concrete prism specimens subjected to accelerated chloride ponding and weekly electrochemical monitoring over eight weeks.

3.2 Materials

3.2.1 Cement

Ordinary Portland Cement (OPC) conforming to KS EAS 18-1 (CEM I, 42.5N), sourced from a locally available Kenyan brand such as Bamburi, Savannah or Mombasa Cement, will be used for all mixes to keep the material variable constant across the experiment.

3.2.2 Fine and Coarse Aggregates

Natural river sand will be used as fine aggregate and crushed granite of maximum nominal size 20 mm as coarse aggregate, both sourced locally and tested for grading, fineness modulus and specific gravity prior to mix design, following standard sieve analysis procedures.

3.2.3 Water

Potable water free of harmful concentrations of chlorides, sulphates or organic matter will be used for both mixing and curing, in line with standard concrete practice.

3.2.4 Reinforcing Steel

Grade 410 mild steel rebar of 12 mm diameter, representative of standard Kenyan construction practice, will be used as the embedded reinforcement in each prism specimen.

3.2.5 Sodium Chloride (NaCl) Solution

A 3-5 percent NaCl solution by mass, representative of seawater chloride concentration, will be used for the accelerated ponding exposure applied to the top surface of each specimen.

3.3 Experimental Design

3.3.1 Cover Depth and Water-Cement Ratio Variables

The experiment varies two independent variables: concrete cover depth (20, 30, 40 and 50 mm) and water-cement ratio (0.40, 0.45, 0.50 and 0.55). This gives a 4 x 4 factorial design, with three replicate specimens per combination, for a total of 48 specimens.

3.3.2 Specimen Configuration Matrix

Table 3.1 below summarises the planned specimen matrix.

Cover Depth (mm)	w/c = 0.40	w/c = 0.45	w/c = 0.50	w/c = 0.55
20	3 specimens	3 specimens	3 specimens	3 specimens
30	3 specimens	3 specimens	3 specimens	3 specimens
40	3 specimens	3 specimens	3 specimens	3 specimens
50	3 specimens	3 specimens	3 specimens	3 specimens

3.4 Specimen Preparation

3.4.1 Mix Design

Mix proportions for each w/c ratio will be determined using the DOE (Department of Environment) mix design method, targeting a consistent workability (slump class S2) across all mixes, with cement content adjusted as necessary to maintain workability at the higher w/c ratios.

3.4.2 Casting and Curing

Reinforced concrete prisms (target dimensions 150 mm x 150 mm x 300 mm) will be cast with a single embedded rebar positioned centrally to achieve the target cover depth for each specimen group, using plastic spacers to maintain position during casting. Specimens will be demoulded after 24 hours and water-cured for 28 days prior to the start of chloride exposure, in accordance with standard curing practice.

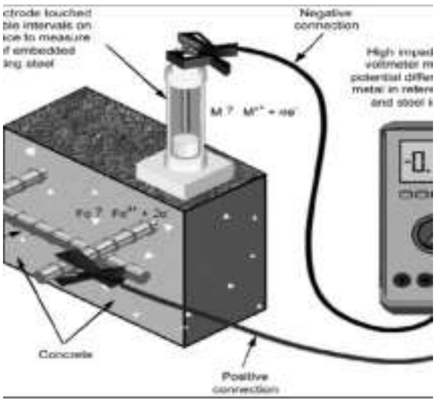
3.5 Accelerated Chloride Exposure (NaCl Ponding)

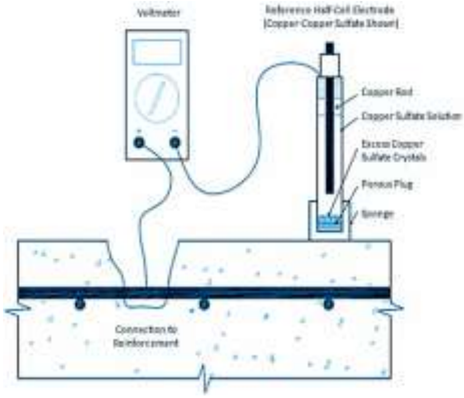
Following curing, a ponding reservoir will be formed on the top surface of each specimen using a sealant dam, and filled with 3-5 percent NaCl solution to a depth of approximately 15 mm. Specimens will undergo weekly wetting-drying cycles (for example, four days wet, three days dry) over an eight-week exposure period to accelerate chloride ingress relative to continuous immersion, while still allowing periodic electrochemical measurement.

3.6 Electrochemical Monitoring

3.6.1 Half-Cell Potential Measurement

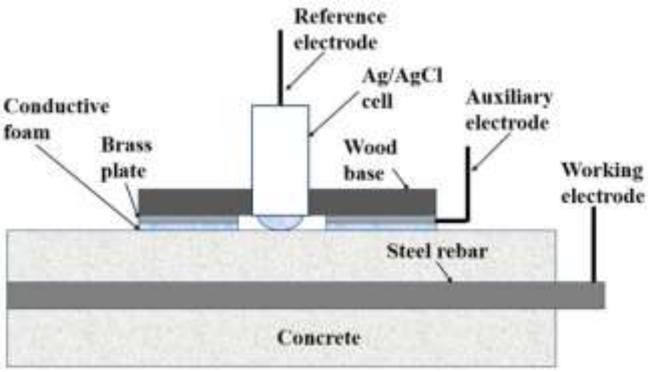
Weekly half-cell potential readings will be taken at the centre of each specimen's exposed surface using a copper-copper sulphate reference electrode connected to a high-impedance voltmeter, following the procedure described in ASTM C876, to track the probability of active corrosion over the exposure period.

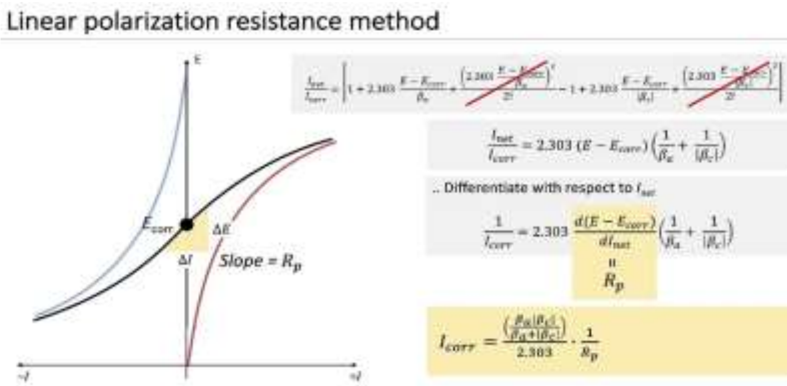


3.6.2 Linear Polarisation Resistance Measurement (Manual)

An industry grade potential meter is very expensive and unobtainable so we will use a manual one. Rather than using a commercial potentiostat, LPR measurements will be obtained manually using a regulated DC bench power supply and two digital multimeters (one connected





in series to measure applied current, one connected in parallel to measure cell potential), together with a stainless-steel mesh counter electrode and the copper-copper sulphate reference electrode used for half-cell potential testing. For each specimen, the open-circuit potential (OCP) will first be allowed to stabilise, after which a small potential perturbation (approximately +/-10 mV to +/-20 mV around OCP) will be applied in a series of discrete steps using the power supply, with the resulting current recorded at each step once it has stabilised.

Plotting the applied potential shift against the recorded current over this small range yields a linear region whose slope is the polarisation resistance (Rp), from which corrosion current density (i_corr) is calculated using the Stern-Geary relationship. This manual stepwise approach is slower per reading than an automated potentiostat scan but requires only standard bench electronics, avoids the cost of dedicated corrosion instrumentation, and gives a quantitative i_corr value consistent with that obtained by automated LPR.

Each manual LPR reading will be repeated at least twice per specimen per week to check consistency, and a subset of readings will be cross-checked against the half-cell potential trend for the same specimen to confirm that both measurements indicate corrosion activity in the same direction over time.

3.7 Derivation of Hydrogen Generation Proxy via Faraday's Law

The weekly manual LPR readings for each specimen are first converted into a polarisation resistance, Rp, taken as the slope of the near-linear region of the applied potential (ΔE) plotted against the resulting current (ΔI) over the ±10-20 mV window around the open-circuit potential, normalised by the exposed rebar surface area to give units of $\Omega\cdot cm^2$. This polarisation resistance is converted into a corrosion current density, i_corr, using the Stern-Geary relationship, where B is the Stern-Geary constant derived from the anodic and cathodic Tafel slopes. Rather than determining Tafel slopes experimentally, which would require a wider potential sweep than the manual LPR set-up is designed for, this study adopts the standard literature values of B = 26 mV for actively corroding steel and B = 52 mV for passive steel, with the appropriate value selected for each specimen and time step based on whether the corresponding half-cell potential reading indicates active corrosion or a passive state.

Polarisation Resistance: $R_p = \frac{\Delta E}{\Delta I}$

Stern-Geary Relationship: $i_{corr} = \frac{B}{R_p}$

Stern-Geary constant from Tafel slopes: $B = \frac{\beta_a \beta_c}{2.303\,(\beta_a + \beta_c)}$

$$B = 26\ \text{mV} \quad \text{(active corrosion)}, \qquad B = 52\ \text{mV} \quad \text{(passive steel)}$$

Once i_corr has been obtained, Faraday's law of electrolysis is applied to convert the measured corrosion current density into an underlying molar rate of electron transfer at the steel surface, where A is the exposed rebar surface area and F is Faraday's constant (96,485 C/mol). This quantity represents the total rate at which electrons generated by anodic dissolution of the steel are being consumed by cathodic reactions at the rebar surface and forms the basis for estimating the rate of hydrogen generation.

Faraday's law, molar rate of electron transfer: $\frac{dn_e}{dt} = \frac{i_{corr} A}{F}$

Because the cathodic reaction at the rebar surface is generally shared between oxygen reduction and hydrogen evolution, and the exact proportion attributable to each cannot be determined without specialised hydrogen permeation cell equipment that falls outside the scope of this study, a conservative worst-case assumption is adopted whereby the entire cathodic current is treated as being consumed by the hydrogen evolution reaction. Under this assumption, the rate of atomic hydrogen generation at the rebar surface is calculated as:

$$\text{Rate}_H = \frac{i_{corr} A}{nF}, \qquad n = 1$$

with n taken as 1 electron per atomic hydrogen produced,

Consistent with the single-electron reduction step that governs the atomic hydrogen available for lattice diffusion and embrittlement.

$$\text{H}^+ + e^- \rightarrow \text{H}_{\text{ads}}$$

This rate is then integrated over the exposure time between successive weekly readings to yield a cumulative hydrogen generation proxy, expressed in moles of atomic hydrogen per specimen per week, for each of the 48 specimens across the eight-week exposure period.

Cumulative hydrogen generation proxy (integrated over exposure time):

$$N_H = \int_{t_0}^{t_0+\Delta t} \text{Rate}_H \; dt \;\approx\; \text{Rate}_H \times \Delta t$$

This proxy is explicitly a relative, upper-bound estimate rather than an absolute measurement of hydrogen concentration in the steel, since the assumption that all cathodic current is consumed by hydrogen evolution is applied uniformly across every specimen regardless of cover depth or w/c ratio. Because the same simplifying assumption is held constant across the full experimental matrix, it does not distort the comparison between specimens, and the resulting proxy remains suitable as the target variable for the Random Forest and SHAP-based sensitivity analysis described in Section 3.8, whose purpose is to identify which design variable dominates hydrogen generation risk rather than to establish its absolute magnitude.

3.8 Data Analysis and Machine Learning Modelling

3.8.1 Dataset Assembly

Data collected across all specimens and time steps (cover depth, w/c ratio, half-cell potential, i_corr, hydrogen generation proxy, and exposure week) will be assembled into a structured dataset of an estimated 300-500 data points, suitable for exploratory statistical and machine learning analysis.

3.8.2 Random Forest Model and SHAP Analysis

A Random Forest regression model will be trained using cover depth, w/c ratio and exposure week as input features and the hydrogen generation proxy as the target variable, implemented in Python using scikit-learn. SHAP (SHapley Additive exPlanations) values will then be computed using the TreeSHAP algorithm to quantify the relative contribution of cover depth and w/c ratio to the model's predictions, and to visualise their interaction effect through a SHAP dependence plot and an interaction-based response surface.

3.8.3 Comparison with Global Benchmarks

Experimentally derived chloride diffusion coefficients and corrosion initiation timelines will be compared against the default parameters and predictions of the Life-365 model, the Bamforth cover-depth relationships, and the Andrade-Alonso corrosion rate classification, in order to quantify any deviation attributable to Kenyan material and climatic conditions.

3.9 Research Timeline

Phase	Activity	Duration
1	Literature review and baseline model calibration	3 weeks
2	Mix design, specimen casting and curing (4x4 factorial, 3 replicates = 48 specimens)	September (4 weeks)
3	Accelerated NaCl ponding with weekly half-cell potential and LPR measurements	Oct & Nov (8 weeks)
4	Faraday's law calculations, dataset assembly, end-of-exposure mass loss validation	Dec (1 week)
5	Random Forest ML modelling with SHAP analysis; response surface visualisation; global benchmark comparison	Dec (2 weeks)
6	Review and Submission	Dec (1 week)

Key equipment required: concrete casting moulds, NaCl ponding tanks, copper-copper sulphate half-cell electrode, a regulated DC bench power supply and two digital multimeters for the manual LPR set-up (in place of a dedicated potentiostat), analytical balance, and vernier callipers. Machine learning analysis will be conducted in Python using scikit-learn, SHAP, and Plotly libraries.

3.10 Evaluation of Methodology

The proposed methodology relies on standard, widely validated electrochemical techniques (ASTM C876 half-cell potential, LPR) and a well-established analytical relationship (Faraday's law) to derive the hydrogen generation proxy, which supports the internal validity of the approach. The 4 x 4 factorial design with triplicate specimens allows both the individual and interactive effects of cover depth and w/c ratio to be estimated, while the accelerated NaCl ponding regime, though not a perfect replica of field exposure, is a widely used and internationally recognised method for compressing multi-year field corrosion processes into a laboratory-scale timeframe. Cross-checking Faraday's law-derived estimates against gravimetric mass loss provides an internal reliability check on the corrosion current data feeding into the hydrogen generation proxy and the subsequent machine learning analysis.

Replacing a commercial potentiostat with a manual DC power supply and multimeter set-up introduces a degree of additional measurement variability, since potential steps and stabilisation times are applied and read manually rather than by an automated scan. This risk is mitigated by taking duplicate readings per specimen per week, applying a consistent stabilisation waiting period before each current reading, and cross-checking the resulting i_corr trend against the independently measured half-cell potential trend for the same specimen; a specimen showing an increasing corrosion current should also show an increasingly negative half-cell potential over the same period. Any specimen for which the two measurements diverge markedly will be flagged and re-measured before being included in the final dataset.

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APPENDICES

Appendix 1: Specimen casting and mix design record sheet (to be completed during Phase 2).

Appendix 2: Weekly half-cell potential and LPR data collection sheet (to be completed during Phase 3).

Appendix 3: Python code listing for Random Forest model training and SHAP analysis (to be completed during Phase 5).